Triangle congruence tests

1

a Yes, they satisfy RHS.

b Unknown, there is not enough information.

c Yes, they satisfy AAS.

d No, they are definitely not congruent.

e Yes, they satisfy SAS.

f Yes, they satisfy RHS.

g No, they are definitely not congruent.

h Yes, they satisfy SSS.

i Yes, they satisfy AAS.

j Yes, they satisfy SSS.

2

a Triangle C and D

b AAS

3

a Triangle B and C

b RHS

4

a Triangle C and D

b SSS

5

a Triangle A and B

b AAS

6

a Triangle B and C

b SAS

7

a Triangle A and C

b RHS

8

a Triangle A and C

b SAS

9

a Triangle B and C

b RHS

10

a Triangle A and D, Triangle B and C

b AAS or ASA

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11 SAS

12

a $\triangle ACD$ and $\triangle BCE$

b SSS

13

a $\triangle JKL$ and $\triangle JMN$

b AAS

14

a $\triangle PRN$ and $\triangle QRM$.

b SSS and SAS.

15 Possible Answers:

$\angle RPQ = \angle RPS, QR = QS, \angle QRP = \angle SRP$

16 Possible Answers:

$JN = MN, \angle JKN = \angle MLN, \angle KJN = \angle LMN$

17 Possible Answers:

$\angle PXQ = \angle RXS, \angle PQX = \angle SRX, \angle PQX = \angle RSX, \angle QPX = \angle SRX,$
$\angle QPX = \angle RSX, PQ = SR$

18

a SAS

b $EF = EH$ Given

$\angle FEG = \angle HED$ Vertically opposite angles

$EG = ED$ Given

19

a SAS

b $\angle AXB = \angle DXC$ Vertically opposite angles

$BX = CX$ Radii of the same circle

$AX = DX$ Radii of the same circle

20

a SAS

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b $AB = CD$ Given

BD is common

$\angle ABD = \angle CDB$ Alternate angles on parallel lines

21

a AAS or SAS

b Example Answer:

$BX = CX$ Given

$\angle ABX = \angle DCX$ Alternate angles on parallel lines

$\angle BXA = \angle CXD$ Vertically opposite angles

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